Aaron McLean

Embedded Software Engineer

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Work Experience

Software & Controls Engineer, Telematics Platform Lead *May 2025 – Present*

MEDATech Engineering, Collingwood, Ontario

- Led the development direction of the machine telemetry platform, owning delivery and support for a production heavy-equipment fleet of 50+ deployed devices
- Owned and operated the platform's AWS stack: loggers on a WireGuard VPN into EC2, nginx terminating TLS in front of a Next.js fleet portal with Cognito auth, DynamoDB for machine-fault events, Timestream and Grafana for customer telemetry, and S3 for log storage
- Deployed an SSRF-safe reverse proxy giving users remote access to on-device dashboards, enforced per-customer Grafana RBAC, and modelled AWS cost scaling per connected logger
- Ran the full lifecycle of every feature, from client requirements and effort/cloud-cost scoping through build, validation, rollout and production monitoring; led a fleet-wide reliability initiative that cut data loss
- Took a first-of-its-kind precision mining system from development to field deployment, owning the operator interface, control system integration, test execution and on-site commissioning
- Led controls for heavy-equipment electrification programs and owned software service for a client's battery-electric rig fleet, including on-site releases across three US states
- Built a production hardware-in-the-loop test system that took engineering out of module verification

Software Developer Intern *Sep 2024 – Dec 2024*

Chimoney (Techstars '23), Toronto, Ontario (Remote)

- Built subaccount payout routing and cross-platform integrations for a distributed payments API (Node.js) on a global payouts platform, in a remote-first engineering team

Education

Carleton University, Ottawa, Ontario *2021 – 2025*

Bachelor of Computer Science (B.C.S.)

- C.J. Mackenzie Scholarship recipient
- Pursued research in quantum communication

Technical Projects

Quantum Blockchain Consensus Protocol

- Designed a quantum-safe blockchain consensus protocol that replaces proof-of-work with quantum-state fidelity checks, implementing quantum hashing and entanglement verification in Python (IBM Qiskit)
- Benchmarked against classical consensus on the Qiskit Aer simulator and published a white paper

ParticleBox: Multithreaded Particle Engine

- Built a real-time particle simulation engine in C++17 with multithreaded updates, spatial hashing and quadtree partitioning, rendered with SDL2

Edge-Optimized Medical Image Classifier

- Trained a CNN for chest X-ray classification to 97.55% accuracy, using mixed-precision inference and GPU acceleration to fit memory-constrained hardware

Technical Skills

Embedded Linux: systemd, C, C++, Python, WireGuard, UDP/TCP logging, OTA updates, IoT telemetry

Software Engineering: AWS (EC2, S3, DynamoDB, Timestream), nginx, TypeScript, Next.js, Node.js, SQL

Mechatronics: CAN (J1939, CANopen), PROFINET, IO-Link, Modbus, HV electrification, HIL testing

Automation: IEC 61131-3, CODESYS, PLCnext, MATLAB/Simulink, PID control, state machines, HMI